Mathematical Foundations of the WCS-GNCCP Molecular Graph Kernel with Hierarchical 2D–3D Alignment

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***Abstract*—We present the mathematical foundations of a hierarchical molecular similarity kernel that combines 2D topological matching with 3D geometric alignment. The 2D stage solves a Weighted Common Subgraph (WCS) problem via Graduated Non-Convexity and Concavity Procedure (GNCCP) with Frank-Wolfe optimization, producing both a similarity score and a bijective atom correspondence. We derive a closed-form structural gradient (Theorem 1), prove a rank-one factorization that eliminates one cubic sub-operation per gradient evaluation (Lemma 1), and establish scale invariance via Frobenius normalization (Proposition 4). The bijective map produced by Stage 1 reduces the 3D alignment (Stage 2) to a single $O(n)$ Kabsch rotation, yielding a hybrid kernel with per-pair complexity dominated by the 2D stage at $O(T_\zeta T_{\mathrm{FW}} n^3)$ — empirically competitive with Random Walk and Optimal Assignment kernels on drug-like molecules.**

I. Problem Formulation

Let G and H be two molecular graphs with M and N atoms, respectively. Their topology is encoded by weighted adjacency matrices $\mathbf{A}_G \in \mathbb{R}^{M\times M}$ and $\mathbf{A}_H \in \mathbb{R}^{N\times N}$, where each entry reflects bond type (single, double, triple, aromatic) and topology (ring membership, conjugation). A cost matrix $\mathbf{C} \in \mathbb{R}^{M\times N}$ encodes pairwise atom-feature dissimilarity. To enforce sparsity and prohibit physically impossible matches, we apply heuristic atom pruning by assigning a sufficiently large penalty ($C_{ij} \to \infty$) for mismatched element types (e.g., pairing Carbon to Oxygen).

Definition 1 (Partial Permutation Polytope)**.** *For an integer $L \le \min(M, N)$, the partial permutation polytope is*

$$\mathcal{P}_L = \big\{\mathbf{X} \in \{0,1\}^{M\times N} \;\big|\; \mathbf{X}\mathbf{1}_N \le \mathbf{1}_M,\; \mathbf{X}^T\mathbf{1}_M \le \mathbf{1}_N,\; \mathbf{1}_M^T\mathbf{X}\mathbf{1}_N = L\big\}, \tag{1}$$

and its convex relaxation $\mathcal{D}_L$ replaces $\{0,1\}^{M\times N}$ with $[0,1]^{M\times N}$. An entry $X_{ij} = 1$ indicates that atom i in G is matched to atom j in H.

Definition 2 (WCS Objective)**.** *The Weighted Common Subgraph (WCS) objective is*

$$F(\mathbf{X}) = \alpha\, f(\mathbf{X}) + (1-\alpha)\,\mathrm{tr}\big(\mathbf{C}^T\mathbf{X}\big), \tag{2}$$

where $\alpha \in [0,1]$ balances structure against atom features,

$$f(\mathbf{X}) = \big\|\mathbf{U}\circ\mathbf{A}_G - \mathbf{X}\mathbf{A}_H\mathbf{X}^T\big\|_F^2 \tag{3}$$

is the structural mismatch term, $\circ$ denotes the Hadamard product, and

$$\mathbf{U} = \mathbf{X}\,\mathbf{1}_{N\times N}\,\mathbf{X}^T \in \mathbb{R}^{M\times M} \tag{4}$$

is the co-matching indicator.

II. Computational Identity for U

The matrix $\mathbf{U}$ in (4) involves the $N\times N$ all-ones matrix, requiring $O(MN^2 + M^2N)$ work. We show that $\mathbf{U}$ admits a rank-one factorization that reduces this to $O(MN + M^2)$.

Lemma 1 (Rank-One Factorization of $\mathbf{U}$)**.** *Let $\mathbf{r} = \mathbf{X}\mathbf{1}_N \in \mathbb{R}^M$ be the row-sum vector. Then*

$$\mathbf{U} = \mathbf{X}\,\mathbf{1}_{N\times N}\,\mathbf{X}^T = \mathbf{r}\,\mathbf{r}^T. \tag{5}$$

Proof. Factor the all-ones matrix as $\mathbf{1}_{N\times N} = \mathbf{1}_N\,\mathbf{1}_N^T$:

$$\begin{aligned}\mathbf{U} &= \mathbf{X}\big(\mathbf{1}_N\,\mathbf{1}_N^T\big)\mathbf{X}^T \\ &= \big(\mathbf{X}\mathbf{1}_N\big)\big(\mathbf{X}\mathbf{1}_N\big)^T = \mathbf{r}\,\mathbf{r}^T.\end{aligned} \tag{6}$$

□

Remark 1. *$\mathbf{U}$ is symmetric PSD with rank one. For binary $\mathbf{X} \in \mathcal{P}_L$ each $r_i \in \{0,1\}$, so $U_{ik} = 1$ iff both atoms i, k are matched.*

III. Gradient of the Structural Term

We derive $\nabla_{\mathbf{X}} f$ for the structural cost (3).

Theorem 1 (Structural Gradient)**.** *Define the residual $\mathbf{R} = \mathbf{U}\circ\mathbf{A}_G - \mathbf{X}\mathbf{A}_H\mathbf{X}^T$. Then*

$$\nabla_{\mathbf{X}} f = 4\big(\mathbf{R}\circ\mathbf{A}_G\big)\mathbf{X}\,\mathbf{1}_{N\times N} - 4\,\mathbf{R}\,\mathbf{X}\,\mathbf{A}_H. \tag{7}$$

Moreover, using Lemma 1:

$$\big(\mathbf{R}\circ\mathbf{A}_G\big)\mathbf{X}\,\mathbf{1}_{N\times N} = \Big[\big(\mathbf{R}\circ\mathbf{A}_G\big)\mathbf{X}\Big]\mathbf{1}_N\mathbf{1}_N^T, \tag{8}$$

a rank-one matrix whose i-th row repeats $\sum_j \big[(\mathbf{R}\circ\mathbf{A}_G)\mathbf{X}\big]_{ij}$.

Proof. Write $f = \mathrm{tr}(\mathbf{R}^T\mathbf{R})$. By the chain rule,

$$\mathrm{d}f = 2\,\mathrm{tr}\big(\mathbf{R}^T\,\mathrm{d}\mathbf{R}\big). \tag{9}$$

Part 1 (Hadamard term). Since $\mathbf{A}_G$ is constant,

$$\mathrm{d}\big(\mathbf{U}\circ\mathbf{A}_G\big) = \big(\mathrm{d}\mathbf{U}\big)\circ\mathbf{A}_G. \tag{10}$$

From $\mathbf{U} = \mathbf{r}\,\mathbf{r}^T$ with $\mathbf{r} = \mathbf{X}\mathbf{1}_N$:

$$\mathrm{d}\mathbf{U} = (\mathrm{d}\mathbf{X}\,\mathbf{1}_N)\,\mathbf{r}^T + \mathbf{r}\,(\mathrm{d}\mathbf{X}\,\mathbf{1}_N)^T. \tag{11}$$

Part 2 (Bilinear term). By the product rule,

$$\mathrm{d}\big(\mathbf{X}\mathbf{A}_H\mathbf{X}^T\big) = (\mathrm{d}\mathbf{X})\,\mathbf{A}_H\,\mathbf{X}^T + \mathbf{X}\,\mathbf{A}_H\,(\mathrm{d}\mathbf{X})^T. \tag{12}$$

Part 3 (Combining). Substitute (10)–(12) into (9) and apply the Hadamard trace identity

$$\mathrm{tr}\big(\mathbf{A}^T(\mathbf{B}\circ\mathbf{C})\big) = \mathrm{tr}\big((\mathbf{A}\circ\mathbf{B})^T\mathbf{C}\big).$$

Collecting terms in $\mathrm{d}\mathbf{X}$: the Hadamard contribution yields $(\mathbf{R}\circ\mathbf{A}_G)\mathbf{X}\mathbf{1}_{N\times N}$ and the bilinear contribution yields $-\mathbf{R}\mathbf{X}\mathbf{A}_H$. Since $\mathbf{R}$, $\mathbf{A}_G$, $\mathbf{U}$, and $\mathbf{X}\mathbf{A}_H\mathbf{X}^T$ are all symmetric, each of the two terms in (11) and (12) contributes equally, producing the factor of 4. □

Remark 2. *A common error is to differentiate $\mathbf{U}\circ\mathbf{A}_G$ as if it were the matrix product $\mathbf{U}\mathbf{A}_G$. The Hadamard derivative is $\mathrm{d}(\mathbf{U}\circ\mathbf{A}_G) = (\mathrm{d}\mathbf{U})\circ\mathbf{A}_G$ (element-wise),* not *$(\mathrm{d}\mathbf{U})\mathbf{A}_G$.*

IV. The GNCCP Continuation Method

Direct minimization of F over $\mathcal{D}_L$ is nonconvex. GNCCP introduces $\zeta\in[-1,1]$ to deform the problem from convex to concave.

Definition 3 (GNCCP Objective)**.** *For $\zeta\in[-1,1]$, define*

$$J_\zeta(\mathbf{X}) = \begin{cases} (1-\zeta)\,F(\mathbf{X}) + \zeta\,\|\mathbf{X}\|_F^2 & \zeta \geq 0, \\ (1+\zeta)\,F(\mathbf{X}) + \zeta\,\|\mathbf{X}\|_F^2 & \zeta < 0. \end{cases} \tag{13}$$

Proposition 1 (Endpoint Properties)**.**

1) *$\zeta = +1$: $J_{+1} = \|\mathbf{X}\|_F^2$ is strictly convex. Its unique minimizer over $\mathcal{D}_L$ is the uniform matrix $\mathbf{X}^{(0)} = \frac{L}{MN}\,\mathbf{1}_{M\times N}$.*
2) *$\zeta = 0$: $J_0 = F(\mathbf{X})$, the original WCS objective.*
3) *$\zeta = -1$: $J_{-1} = -\|\mathbf{X}\|_F^2$ is strictly concave. Its minimizers over $\mathcal{D}_L$ are vertices, i.e., partial permutation matrices in $\mathcal{P}_L$.*

Proof. (1) $\|\mathbf{X}\|_F^2 = \sum_{ij} X_{ij}^2$ is strictly convex. The unique minimizer distributes mass uniformly: $X_{ij} = L/(MN)$ satisfies all constraints and minimizes the sum of squares by Schur convexity.

(2) Direct substitution.

(3) $-\|\mathbf{X}\|_F^2$ is strictly concave. A strictly concave function attains its minimum over a compact convex polytope at a vertex. The vertices of $\mathcal{D}_L$ are exactly the elements of $\mathcal{P}_L$. □

Proposition 2 (Gradient of J_ζ)**.**

$$\nabla_{\mathbf{X}} J_\zeta = \begin{cases} (1-\zeta)\,\nabla_{\mathbf{X}} F + 2\zeta\,\mathbf{X} & \zeta \geq 0, \\ (1+\zeta)\,\nabla_{\mathbf{X}} F + 2\zeta\,\mathbf{X} & \zeta < 0, \end{cases} \tag{14}$$

where $\nabla_{\mathbf{X}} F = \alpha\,\nabla_{\mathbf{X}} f + (1-\alpha)\,\mathbf{C}$ and $\nabla_{\mathbf{X}} f$ is given by Theorem 1.

Proof. $\nabla\|\mathbf{X}\|_F^2 = 2\mathbf{X}$. Apply linearity to (13). □

V. Frank-Wolfe Optimization

For each fixed ζ we solve $\min_{\mathbf{X}\in\mathcal{D}_L} J_\zeta(\mathbf{X})$ via Frank-Wolfe (conditional gradient).

Definition 4 (Linear Minimization Oracle)**.** *At iterate $\mathbf{X}_t$, the FW direction is*

$$\mathbf{Y}_t = \arg\min_{\mathbf{Y}\in\mathcal{D}_L} \big\langle \nabla J_\zeta(\mathbf{X}_t),\, \mathbf{Y}\big\rangle. \tag{15}$$

Since the minimum of a linear function over a polytope is attained at a vertex, $\mathbf{Y}_t\in\mathcal{P}_L$. The optimal vertex is found by the Hungarian algorithm in $O\big(\min(M,N)^3\big)$ time.

Theorem 2 (Frank-Wolfe Duality Gap)**.** *Let J_ζ be convex on $\mathcal{D}_L$ and let $\mathbf{X}^* = \arg\min_{\mathcal{D}_L} J_\zeta$. The Frank-Wolfe gap*

$$g_t = \big\langle \nabla J_\zeta(\mathbf{X}_t),\ \mathbf{X}_t - \mathbf{Y}_t\big\rangle \tag{16}$$

satisfies: (i) $g_t \geq 0$; (ii) $g_t \geq J_\zeta(\mathbf{X}_t) - J_\zeta(\mathbf{X}^*)$; (iii) $g_t = 0 \Leftrightarrow \mathbf{X}_t$ *is optimal.*

Proof. (i) $\mathbf{Y}_t$ minimizes the linear form over $\mathcal{D}_L$ and $\mathbf{X}_t\in\mathcal{D}_L$, so $\langle\nabla J_\zeta, \mathbf{Y}_t\rangle \leq \langle\nabla J_\zeta, \mathbf{X}_t\rangle$, giving $g_t \geq 0$.

(ii) By convexity:

$$J_\zeta(\mathbf{X}^*) \geq J_\zeta(\mathbf{X}_t) + \big\langle \nabla J_\zeta(\mathbf{X}_t),\, \mathbf{X}^* - \mathbf{X}_t\big\rangle. \tag{17}$$

Since $\mathbf{Y}_t$ is the minimizer and $\mathbf{X}^*\in\mathcal{D}_L$:

$$\big\langle \nabla J_\zeta(\mathbf{X}_t),\, \mathbf{X}^* - \mathbf{X}_t\big\rangle \geq \big\langle \nabla J_\zeta(\mathbf{X}_t),\, \mathbf{Y}_t - \mathbf{X}_t\big\rangle = -g_t. \tag{18}$$

Combining (17)–(18): $g_t \geq J_\zeta(\mathbf{X}_t) - J_\zeta(\mathbf{X}^*)$.

(iii) If $\mathbf{X}_t$ is optimal, first-order optimality gives $\langle\nabla J_\zeta, \mathbf{Y} - \mathbf{X}_t\rangle \geq 0\ \forall\,\mathbf{Y}\in\mathcal{D}_L$, so $g_t = 0$. Conversely, $g_t = 0$ with (ii) gives $J_\zeta(\mathbf{X}_t) \leq J_\zeta(\mathbf{X}^*)$. □

Remark 3. *When J_ζ is nonconvex ($\zeta<0$), the gap* (16) *is no longer a certified suboptimality bound, but $g_t = 0$ still implies $\mathbf{X}_t$ is a stationary point. We use $g_t < \varepsilon$ as early termination for all ζ.*

VI. Kernel Properties

Proposition 3 (Exponential Graph Similarity Kernel)**.** *The exponential mapping*

$$k(G,H) = \exp\Big(-\frac{\gamma\,F^*_{GH}}{L}\Big) \tag{19}$$

where $\gamma > 0$ is a bandwidth parameter and F^*_{GH} is the optimal WCS objective, defines a discriminative similarity measure satisfying:

1) *$k(G,H)\in(0,1]$ for all graph pairs;*
2) *$k(G,G) = 1$ when G is matched to itself;*
3) *k is monotonically decreasing in F^*_{GH}.*

Remark 4. *The kernel* (19) *is generally* not *positive semi-definite in the Mercer sense, since optimal-assignment and graph-edit distances do not satisfy conditional negative definiteness (cf. Schoenberg [1]). This is not a limitation for distance-based classifiers such as k-NN (which rely on ranking, not inner products) or for SVM with eigenvalue ridge correction ($\mathbf{K}_{psd} = \mathbf{K} + \mu\mathbf{I}$, $\mu \geq |\lambda_{\min}|$).*

Remark 5. *γ controls bandwidth: small γ spreads kernel values (all pairs appear similar), large γ concentrates mass on near-identical pairs. We default to $\gamma = 0.5$ and normalize by $\max(L, 1)$ for size invariance.*

VII. Scale Invariance

Proposition 4 (Frobenius Normalization)**.** *Let $\hat{\mathbf{A}}_G = \mathbf{A}_G/\|\mathbf{A}_G\|_F$ and $\hat{\mathbf{A}}_H = \mathbf{A}_H/\|\mathbf{A}_H\|_F$. Then*

$$\hat{f}(\mathbf{X}) = \left\|\mathbf{U} \circ \hat{\mathbf{A}}_G - \mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T\right\|_F^2 \leq 2 \tag{20}$$

for all $\mathbf{X} \in \mathcal{D}_L$, independently of the number of bonds.

Proof. By the identity $\|\mathbf{A}-\mathbf{B}\|_F^2 = \|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2 - 2\langle \mathbf{A}, \mathbf{B}\rangle$, it suffices to bound each squared norm.

For the Hadamard term, using $\|\mathbf{A}\circ\mathbf{B}\|_F \leq \|\mathbf{A}\|_{\max}\|\mathbf{B}\|_F$ with $U_{ik} = r_i r_k \leq 1$ for $\mathbf{X} \in \mathcal{D}_L$:

$$\left\|\mathbf{U} \circ \hat{\mathbf{A}}_G\right\|_F \leq 1 \cdot \|\hat{\mathbf{A}}_G\|_F = 1. \tag{21}$$

For the bilinear term, since $\|\mathbf{X}\|_2 \leq 1$ for sub-stochastic $\mathbf{X}$:

$$\left\|\mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T\right\|_F \leq \|\mathbf{X}\|_2^2\,\|\hat{\mathbf{A}}_H\|_F \leq 1. \tag{22}$$

Therefore $\hat{f} \leq \|\mathbf{U}\circ\hat{\mathbf{A}}_G\|_F^2 + \|\mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T\|_F^2 \leq 2$. □

Remark 6. *Without normalization, $f(\mathbf{X})$ scales with $\|\mathbf{A}_G\|_F^2 + \|\mathbf{A}_H\|_F^2$, which grows linearly in the number of bonds. For 30 bonds at weight $\sim$1.5, $\|\mathbf{A}_G\|_F \approx 9.5$, making the structural cost $\sim$90$\times$ larger than the appearance cost $\mathrm{tr}(\mathbf{C}^T\mathbf{X}) \in [0, L]$. Normalization restores the balance intended by α.*

VIII. Projection onto $\mathcal{D}_L$

Proposition 5 (Sinkhorn-Knopp Projection)**.** *Let $\mathbf{X} \in \mathbb{R}_{\geq 0}^{M\times N}$ with $\|\mathbf{X}\|_1 > 0$. The iteration*

1. ***Scale**: $\mathbf{X} \leftarrow \mathbf{X} \cdot (L/\|\mathbf{X}\|_1)$.*
2. ***Row cap**: if $(\mathbf{X}\mathbf{1}_N)_i > 1$, divide row i by $(\mathbf{X}\mathbf{1}_N)_i$.*
3. ***Col cap**: if $(\mathbf{X}^T\mathbf{1}_M)_j > 1$, divide col j by $(\mathbf{X}^T\mathbf{1}_M)_j$.*
4. *Repeat 1–3 until convergence.*

converges to a feasible point in $\mathcal{D}_L$ when $L \leq \min(M, N)$.

Proof sketch. Steps 2–3 are alternating projections onto $\mathcal{R} = \{\mathbf{X} \geq 0 : \mathbf{X}\mathbf{1} \leq \mathbf{1}\}$ and $\mathcal{C} = \{\mathbf{X} \geq 0 : \mathbf{X}^T\mathbf{1} \leq \mathbf{1}\}$, both convex and closed. Step 1 enforces $\|\mathbf{X}\|_1 = L$. By Dykstra's theorem, this converges when the intersection is non-empty. The uniform matrix $\frac{L}{MN}\mathbf{1}_{M\times N}$ lies in all three constraint sets, so the intersection is non-empty. □

IX. Computational Complexity Analysis

We provide an honest, operation-level complexity analysis of each Frank-Wolfe iteration.

Theorem 3 (Per-Iteration Complexity)**.** *Let M, N denote the atom counts and let d be the maximum vertex degree of the molecular graphs. One Frank-Wolfe iteration of Algorithm 1 has complexity*

$$\Theta\big(\underbrace{M^2N}_{\mathbf{X}\mathbf{A}_H\mathbf{X}^T} + \underbrace{\min(M,N)^3}_{\textit{Hungarian}}\big). \tag{23}$$

For $M \approx N$, this simplifies to $\Theta(n^3)$.

Proof. We enumerate the dominant operations per iteration:

(a) Row sums: $\mathbf{r} = \mathbf{X}\mathbf{1}_N$ costs $O(MN)$.

(b) Co-matching indicator (Lemma 1): $\mathbf{U} = \mathbf{r}\mathbf{r}^T$ costs $O(M^2)$. Without Lemma 1, the naïve computation via (4) costs $O(MN^2 + M^2N)$.

(c) Bilinear form: $\mathbf{P} = \mathbf{X}\mathbf{A}_H$ costs $O(M \cdot \mathrm{nnz}(\mathbf{A}_H))$; for molecular graphs with bounded degree d, $\mathrm{nnz}(\mathbf{A}_H) = O(dN)$, so $\mathbf{P}$ costs $O(MdN) \approx O(MN)$. However, $\mathbf{P}\mathbf{X}^T$ is an $(M \times N)(N \times M)$ product costing $O(M^2N)$. This is irreducible for dense $\mathbf{X}$.

(d) Gradient terms: $(\mathbf{R}\circ\mathbf{A}_G)\mathbf{X}$ costs $O(\mathrm{nnz}(\mathbf{A}_G)\cdot N) = O(dMN)$; $\mathbf{R}\mathbf{X}\mathbf{A}_H$ costs $O(M^2N + MdN)$.

(e) Hungarian algorithm: $O(\min(M,N)^3)$; this is tight (Kuhn–Munkres).

(f) Projection: $O(MN)$ per Sinkhorn iteration, bounded at 20 iterations.

The dominant terms are (c) and (e), giving (23). □

Corollary 1 (Reduction from Lemma 1)**.** *The rank-one factorization eliminates one $O(MN^2 + M^2N)$ operation per gradient evaluation, reducing it to $O(MN + M^2)$. For $M = N = n$, this is a reduction from $O(n^3)$ to $O(n^2)$ for the $\mathbf{U}$ computation alone. The overall per-iteration complexity remains $\Theta(n^3)$ due to the bilinear form (c) and Hungarian assignment (e).*

Corollary 2 (Total Per-Pair Complexity)**.** *With T_ζ GNCCP outer iterations and T_{FW} Frank-Wolfe inner iterations, the total per-pair cost is*

$$O\big(T_\zeta\, T_{\mathrm{FW}}\, n^3\big). \tag{24}$$

In practice, early termination via the FW duality gap (Theorem 2) reduces the effective iteration count well below the worst case.

Remark 7 (Practical Competitiveness)**.** *For drug-like molecules ($n \leq 50$), the cubic cost per iteration is sub-millisecond. With $(T_\zeta, T_{\mathrm{FW}}) = (4, 4)$ and hydrogen removal ($n \approx 30$ heavy atoms), the measured per-pair time is $\sim$6 ms on CPU — competitive with Random Walk ($O(n^3)$, $\sim$10–30 ms) and Optimal Assignment kernels ($O(n^3)$, $\sim$10–50 ms), while producing a richer output (bijective atom map in addition to similarity score).*

X. Stage 2: 3D Alignment via Kabsch Rotation

The bijective atom correspondence $\mathbf{X}^*$ from Stage 1 induces a set of L matched atom pairs $\mathcal{M} = \{(i, j) : X^*_{ij} = 1\}$. We exploit this mapping to perform 3D alignment in $O(L)$ time.

Definition 5 (Matched Coordinate Sets)**.** *Let $\mathbf{p}_1, \ldots, \mathbf{p}_L \in \mathbb{R}^3$ and $\mathbf{q}_1, \ldots, \mathbf{q}_L \in \mathbb{R}^3$ be the 3D coordinates of the matched atoms in molecules G and H respectively, ordered according to $\mathcal{M}$. Define centered coordinates $\bar{\mathbf{p}}_l = \mathbf{p}_l - \boldsymbol{\mu}_P$ and $\bar{\mathbf{q}}_l = \mathbf{q}_l - \boldsymbol{\mu}_Q$, where $\boldsymbol{\mu}_P = \frac{1}{L}\sum_l \mathbf{p}_l$ and $\boldsymbol{\mu}_Q = \frac{1}{L}\sum_l \mathbf{q}_l$.*

Theorem 4 (Kabsch Alignment)**.** *The rotation matrix* $\mathbf{R}^* \in \mathrm{SO}(3)$ *that minimizes the RMSD*

$$\mathrm{RMSD}^2 = \frac{1}{L}\sum_{l=1}^{L} \|\mathbf{R}\bar{\mathbf{p}}_l - \bar{\mathbf{q}}_l\|^2 \tag{25}$$

is given by

$$\mathbf{R}^* = \mathbf{V}\begin{pmatrix} 1 & & \\ & 1 & \\ & & \det(\mathbf{V}\mathbf{W}^T) \end{pmatrix}\mathbf{W}^T, \tag{26}$$

where $\mathbf{H} = \bar{\mathbf{P}}^T\bar{\mathbf{Q}} \in \mathbb{R}^{3\times 3}$ *is the cross-covariance matrix and* $\mathbf{H} = \mathbf{W}\mathbf{\Sigma}\mathbf{V}^T$ *is its SVD.*

Proof. Standard; see Kabsch (1976). The key observation is that $\mathbf{H}$ is always 3×3 regardless of L, so its SVD costs $O(1)$. The total cost is dominated by forming $\mathbf{H}$, which requires $O(L)$ vector operations. □

Corollary 3 (Stage 2 Complexity)**.** *Given the atom map* $\mathcal{M}$ *from Stage 1, the 3D alignment and RMSD computation cost* $O(L) = O(n)$.

Definition 6 (Hybrid Kernel)**.** *The two-stage similarity measure combines 2D topological and 3D geometric information:*

$$k_{\textit{hybrid}}(G,H) = (1-\beta)\,k_{\textit{2D}}(G,H) + \beta\,k_{\textit{3D}}(G,H), \tag{27}$$

where $k_{\textit{2D}} = \exp(-\gamma_{\textit{2D}}F^*/L)$ *is the topological kernel from* (19)*,* $k_{\textit{3D}} = \exp(-\gamma_{\textit{3D}}\cdot\mathrm{RMSD})$ *is the geometric kernel, and* $\beta \in [0,1]$ *is cross-validated per target.*

Remark 8 (Overall Pipeline Complexity)**.** *The total per-pair cost of the hierarchical pipeline is*

$$\underbrace{O(T_\zeta T_{\mathrm{FW}} n^3)}_{\textit{Stage 1: 2D kernel}} + \underbrace{O(n)}_{\textit{Stage 2: 3D alignment}} = O(T_\zeta T_{\mathrm{FW}} n^3), \tag{28}$$

completely dominated by Stage 1. The 2D stage serves as a search-space reduction gate: only candidates with $k_{\textit{2D}} > \tau$ *(a tunable threshold) proceed to Stage 2. For a library of* P *molecules with filtration rate* ϕ*, the total screening cost is*

$$\underbrace{O\left(\binom{P}{2} T_\zeta T_{\mathrm{FW}} n^3\right)}_{\textit{2D pairwise kernel matrix}} + \underbrace{O(\phi P^2 n)}_{\textit{3D alignment of filtered pairs}}. \tag{29}$$

With aggressive filtering ($\phi \ll 1$*), the 3D cost becomes negligible.*

XI. Relabeling Invariance

Proposition 6 (Relabeling Invariance)**.** *Let* $\mathbf{P} \in \{0,1\}^{M\times M}$ *and* $\mathbf{Q} \in \{0,1\}^{N\times N}$ *be permutation matrices, and let* $\mathbf{A}'_G = \mathbf{P}\mathbf{A}_G\mathbf{P}^T$*,* $\mathbf{A}'_H = \mathbf{Q}\mathbf{A}_H\mathbf{Q}^T$*. Then for any* $\mathbf{X} \in \mathcal{D}_L$*,*

$$F\big(\mathbf{P}\mathbf{X}\mathbf{Q}^T;\, \mathbf{A}'_G, \mathbf{A}'_H, \mathbf{C}'\big) = F\big(\mathbf{X};\, \mathbf{A}_G, \mathbf{A}_H, \mathbf{C}\big), \tag{30}$$

where $\mathbf{C}' = \mathbf{P}\mathbf{C}\mathbf{Q}^T$.

Proof. Let $\mathbf{X}' = \mathbf{P}\mathbf{X}\mathbf{Q}^T$. Then $\mathbf{X}'\mathbf{A}'_H(\mathbf{X}')^T = \mathbf{P}\mathbf{X}\mathbf{A}_H\mathbf{X}^T\mathbf{P}^T$ and $\mathbf{U}'\circ\mathbf{A}'_G = \mathbf{P}(\mathbf{U}\circ\mathbf{A}_G)\mathbf{P}^T$. The Frobenius norm is unitarily invariant: $\|\mathbf{P}\mathbf{M}\mathbf{P}^T\|_F = \|\mathbf{M}\|_F$. The trace term satisfies $\mathrm{tr}((\mathbf{C}')^T\mathbf{X}') = \mathrm{tr}(\mathbf{C}^T\mathbf{X})$ by the cyclic property of trace. □

XII. Full Algorithm

Algorithm 1 WCS-GNCCP with Frank-Wolfe

Require: $\mathbf{A}_G, \mathbf{A}_H$; cost $\mathbf{C}$; size L; weights α, γ
Ensure: Similarity score $k \in (0,1]$ and atom map $\mathcal{M}$

$\hat{\mathbf{A}}_G \leftarrow \mathbf{A}_G/\|\mathbf{A}_G\|_F$; $\hat{\mathbf{A}}_H \leftarrow \mathbf{A}_H/\|\mathbf{A}_H\|_F$ {Prop. 4}
$\hat{\mathbf{C}} \leftarrow (\mathbf{C} - C_{\min})/(C_{\max} - C_{\min})$
$\mathbf{X} \leftarrow \frac{L}{MN}\mathbf{1}_{M\times N}$ {Prop. 1(1)}
for $\zeta = 1, 1-\Delta\zeta, \ldots, -1$ **do**
  for $t = 1, \ldots, T_{\mathrm{FW}}$ **do**
    $\nabla \leftarrow \nabla_{\mathbf{X}} J_\zeta$ {Prop. 2}
    $\mathbf{Y} \leftarrow \mathrm{Hungarian}(\nabla)$ {Def. 4}
    $g \leftarrow \langle \nabla, \mathbf{X} - \mathbf{Y}\rangle$ {Thm. 2}
    if $g < \varepsilon$ **then**
      break
    end if
    $\lambda \leftarrow \frac{2}{t+2}$ {Standard FW step decay}
    $\mathbf{X} \leftarrow \mathrm{Sinkhorn}\big(\mathbf{X} + \lambda(\mathbf{Y} - \mathbf{X}),\, L\big)$ {Prop. 5}
  end for
end for
$\mathbf{X}^* \leftarrow \mathrm{Hungarian}(-\mathbf{X})$ {discretize}
$\mathcal{M} \leftarrow \{(i,j) : X^*_{ij} = 1\}$ {atom map}
$F^* \leftarrow F(\mathbf{X}^*)$ {Def. 2}
return $k = \exp(-\gamma F^*/L)$, $\mathcal{M}$ {Prop. 3}

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